

Small Open Multimodal LLMs as Industrial Inspectors: Robustness, Deployability and Detector Guidance on MMAD

(Draft v1 — work in progress)

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Abstract—Closed commercial multimodal large language models (MLLMs) lead the MMAD industrial-anomaly benchmark, but factories rarely can send production images to a cloud API. We ask how far *small, open-weight* MLLMs (0.26–8.2 B parameters) running on a single 8 GB laptop GPU can go, how they fail under factory-floor image corruption, and whether a classical anomaly detector can guide them. On a stratified 2,500-question MMAD sample, zero-shot Qwen3-VL-2B reaches 71.2 % accuracy ($\kappa=0.615$, 0.16 s/question) and Qwen3-VL-8B in 4-bit NF4 reaches 72.2 %; under MMAD’s own seven-task protocol the 8B model scores 70.5, 0.2 points below InternVL2-76B and 4.4 below GPT-4o. Across ten open models, defect localisation separates vision-encoder families more than parameter counts. A 5,000-image corruption study (25,000 inferences) shows that overall accuracy hides large per-subtask effects: severity-4 motion blur costs 3.8 pp overall but 19.4 pp on object classification, and training-free image restoration does not recover it. A first detector-guided pipeline (PatchCore box drawn on the image, “Arm B”) lowered accuracy by 3.6 pp (McNemar $p=0.08$, n.s.); we trace this to a calibration flaw that made the detector fire on 92.6 % of images, and to a single-dataset sample. To study Arm B properly we built an open, node-based research lab in which the detector (PatchCore, WinCLIP, an oracle, null controls), visual prompt, text prompt, MLLM and answer parser are swappable, and every configuration is evaluated with paired statistics against ground-truth masks. This draft reports completed work and lists the experiments that remain.

Index Terms—industrial anomaly detection, multimodal large language models, robustness, visual prompting, PatchCore, WinCLIP, edge deployment, MMAD

I. INTRODUCTION

Automated visual inspection needs more than a defect/no-defect flag: an operator wants to know *what* the defect is, *where* it is and *whether it matters*. MMAD [1] measures exactly this with 39,672 multiple-choice questions over 8,366 images from MVTec-AD [2], MVTec-LOCO [3], VisA [4] and GoodsAD [5]. Its leaderboard is led by GPT-4o (74.9, 1-shot) and the best open model it reports is InternVL2-76B (70.8). Neither is practical on a production line: API models move proprietary images off-site and change silently, and 76 B-parameter models do not fit edge hardware.

This work evaluates the regime a small manufacturer can actually deploy: open-weight models of 0.26–8.2 B parameters

on one RTX 4060 Laptop GPU (8.19 GB). We extend MMAD along three axes it does not cover: robustness to realistic imaging corruptions, the deployment envelope (memory, latency), and *detector guidance*, i.e. letting a classical anomaly detector tell the MLLM where to look (Arm B, Fig. 1).

Contributions (completed).

- 1) A reproducible zero-shot baseline of Qwen3-VL-2B on 2,500 stratified MMAD questions, plus a 1-shot run with MMAD’s official evaluation code on 18,651 questions (Sec. IV-A).
- 2) A ten-model comparison on the same questions, mapped onto MMAD’s seven-task protocol, with memory and throughput (Sec. IV-B).
- 3) A 25,000-inference corruption and test-time-restoration study (Sec. IV-C).
- 4) A first Arm B experiment and a diagnosis of why it failed (Sec. IV-D).
- 5) *Arm-B Lab*, an open node-graph research environment for detector-guided MLLMs with built-in paired evaluation (Sec. V).

Still to do (Sec. VII): the controlled Arm B study that the lab was built for, a complete official-protocol run, and robustness of Arm B itself. Table I summarises the state of each phase.

II. RELATED WORK

MLLMs for industrial inspection. MMAD [1] defines seven inspection subtasks and evaluates 1-shot MLLMs with a normal template image. AnomalyGPT [6] fine-tunes an LVLM with a learned anomaly decoder and scores 36.5 on MMAD, below general-purpose models, which suggests general visual-language competence matters more than narrow anomaly training for MMAD’s question types.

Unsupervised anomaly detection. PatchCore [7] compares mid-level patch features with a coreset memory of normal patches and gives near-perfect localisation on MVTec-AD. WinCLIP [8] scores image windows against “normal/damaged object” prompts in CLIP [9] space, zero-shot or with a few normal shots (WinCLIP+). Both produce heatmaps but no

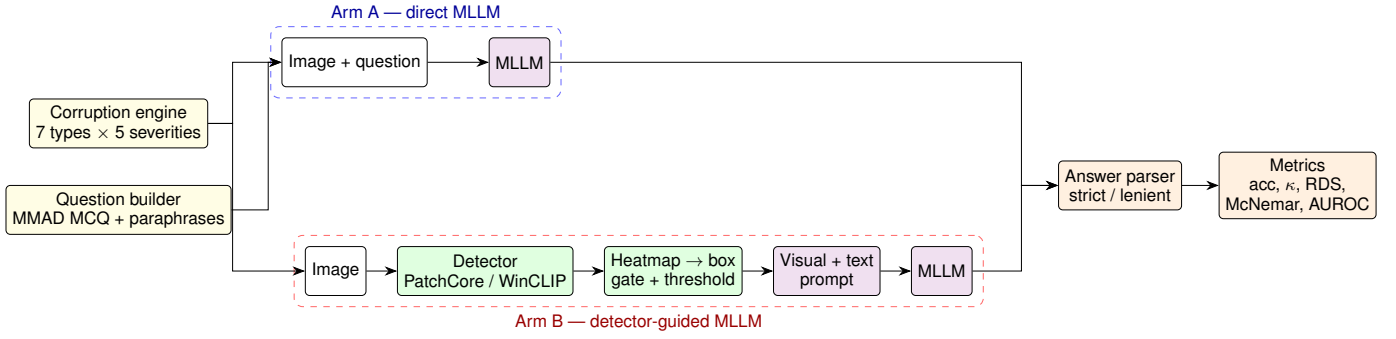


Fig. 1. Study design. Arm A gives the MLLM the image and question directly. Arm B first runs an unsupervised anomaly detector, converts its heatmap into a box, and shows the MLLM where the detector points before it answers. Both arms receive identical (optionally corrupted) images and questions, so they can be compared question by question.

TABLE I
STATUS OF THE STUDY (SEPTEMBER 2026).

Phase	Content	Status
0	Pipeline smoke test (Qwen3-VL-2B)	complete
1	Clean zero-shot baseline, $N=2,500$	complete
1b	Official MMAD code, 1-shot	partial (45% of images)
2	Corruptions: 5k study + 7×5 sweep	complete; sweep underpowered
3	Test-time restoration & prompting	complete
4	Ten-model zoo + deployment envelope	complete
5	Arm B pilot (PatchCore box)	complete; confounded
6	Arm-B Lab (tooling)	built, validated
6+	Controlled Arm B experiments	to do

explanation, which is the complementarity Arm B tries to exploit.

Visual prompting. Drawing a red circle steers CLIP’s attention [10], and Set-of-Mark [11] overlays region marks to ground GPT-4V. Whether such marks help or *mislead* small MLLMs on fine-grained defects is the open question for Arm B.

Robustness. Common-corruption benchmarks [12] show large accuracy drops for vision models; MMAD evaluates only clean captures.

Small open MLLMs. We evaluate Qwen2-VL/2.5-VL/3-VL [13]–[15], SmolVLM [16], Gemma 4 [17], Moon-dream2 [18] and PaliGemma 2 [19]; larger models are run in 4-bit NF4 [20].

III. EXPERIMENTAL SETUP

Data. MMAD’s 8,366 images and 39,672 questions carry nine subtask labels. Following the MMAD paper, *Object Analysis*, *Object Structure* and *Object Details* are folded into one “Object Analysis” column when reporting the seven-task average, and the seven-task average is the unweighted column mean (this reproduces 20 of the 21 printed MMAD rows exactly). Anomaly discrimination is reported as MMAD’s balanced accuracy (mean of normal- and abnormal-class accuracy).

Sampling. Unless stated otherwise, questions are drawn stratified by subtask (equal share per subtask, remainder at random) with seed 42. Phase 1 and Phase 4 use $N=2,500$; the robustness study uses 5,000 questions.

Inference. Zero-shot, greedy decoding, 4–8 new tokens, the question and lettered options followed by “Answer with the letter of the correct option...”. Qwen images are limited to $512 \times 28 \times 28$ pixels. Hardware: RTX 4060 Laptop (8.19 GB), PyTorch 2.6 / CUDA 12.4.

Metrics. Accuracy with Wilson 95 % intervals [21]; Cohen’s κ [22] between predicted and true letters; the robustness degradation slope $RDS(s) = (Acc_{clean} - Acc_s)/s$ in points per severity level; paired comparisons with the exact McNemar test [23]; for detectors, image AUROC, pixel AUROC and box IoU against MMAD’s ground-truth masks. At $n \approx 278$ per subtask the 95 % half-width is about ± 5.8 pp, and at $n=2,500$ about ± 1.8 pp; smaller differences are not claimed as effects.

IV. RESULTS

A. Clean baseline (Phases 0–1)

Qwen3-VL-2B (FP16; 4.26 GB allocated after loading, 4.82 GB peak) answers 1,780 of 2,500 questions correctly: 71.2 % [69.4, 72.9], $\kappa=0.615$, 0.16 s per question (6.2 questions/s), with 100 % of outputs parseable. Table II and Fig. 2 show the profile: object-level questions are easy (Object Classification 94.9 %), while the two questions an inspector cares most about, *which defect* (Defect Classification 40.8 %) and *where* (Defect Localization 47.5 %), are the weakest. MVTec-LOCO, whose anomalies are logical rather than visual, is the hardest source (61.6 %).

With MMAD’s official evaluation code in its 1-shot setting (one normal template), Qwen3-VL-2B reaches 70.8 % on 18,651 questions from 3,800 images (MVTec-AD images via DS-MVTec 78.3 %, VisA 75.8 %, MVTec-LOCO 60.9 %). GoodsAD is not yet covered, so this is not a full-benchmark number **[TODO: finish GoodsAD]**.

B. Model zoo and deployment envelope (Phase 4)

Table III evaluates ten open models on the same 2,500 questions (Phase 1’s draw differs from the Phase-4 draw by four Anomaly Detection questions). Three observations:

TABLE II
QWEN3-VL-2B ZERO-SHOT, $N=2,500$ (PHASE 1).

Subtask	n	Acc	Source	n	Acc
Object Classif.	277	94.9	DS-MVTec	497	77.5
Object Analysis	277	82.3	VisA	787	74.6
Defect Analysis	278	82.0	GoodsAD	716	69.8
Object Structure	277	81.2	MVTec-LOCO	500	61.6
Object Details	277	79.8			
Defect Descr.	277	71.1			
Anomaly Det.	282	61.3			
Defect Local.	278	47.5			
Defect Classif.	277	40.8			
All	2500	71.2			

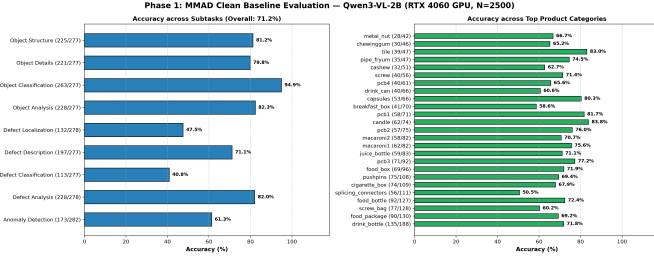


Fig. 2. Phase 1 accuracy by subtask (Qwen3-VL-2B, zero-shot, $N=2,500$).

Small open models reach the upper half of the MMAD leaderboard. Under MMAD’s seven-task protocol (Fig. 3), Qwen3-VL-8B (4-bit) scores 70.5 and Qwen3-VL-2B 68.4, above every open model ≤ 34 B in the MMAD paper and on par with Gemini-1.5-flash (68.9) and Claude-3.5-sonnet (68.4). The 8B and 2B models are not separable from each other at this sample size (72.2 vs 71.2 on nine tasks). These comparisons are 0-shot against the paper’s 1-shot; MMAD’s own 0/1-shot ablation moves models by -1.3 to $+1.9$ points (Fig. 4), so differences below about two points are not meaningful.

Defect localisation separates encoder families. On Defect Localization (Fig. 5), Qwen dynamic-resolution encoders score 47.5–56.0 and Moondream2 59.2, while both Gemma 4 models (31.4, 26.4) and SmolVLM (33.2, 29.6) sit near the 25% chance level; at matched 4.4B scale Qwen3-VL-4B beats Gemma-4-E4B by 20.2 pp. Qwen3-VL-8B (56.0) and Moondream2 (59.2) are statistically indistinguishable from GPT-4o (55.6) at $n \approx 278$ (± 5.8 pp).

Deployability is not monotone in size (Fig. 6). The fastest model (Qwen3-VL-2B, 6.2 q/s) is also among the most accurate; NF4 quantisation puts Qwen3-VL-4B in 3.1 GB. Below 1 B parameters accuracy collapses (SmolVLM-500M 45.7%, SmolVLM-256M 32.6%). PaliGemma 2 (8.2%) is a *pre-trained*, not instruction-tuned, checkpoint that does not produce option letters; it is a format failure rather than a capability measurement and should be dropped or re-run with its mix checkpoint [TODO: decide].

C. Robustness to factory-floor corruptions (Phases 2–3)

Large-scale study. Qwen3-VL-2B was evaluated on 5,000 questions in five conditions (25,000 inferences): clean,

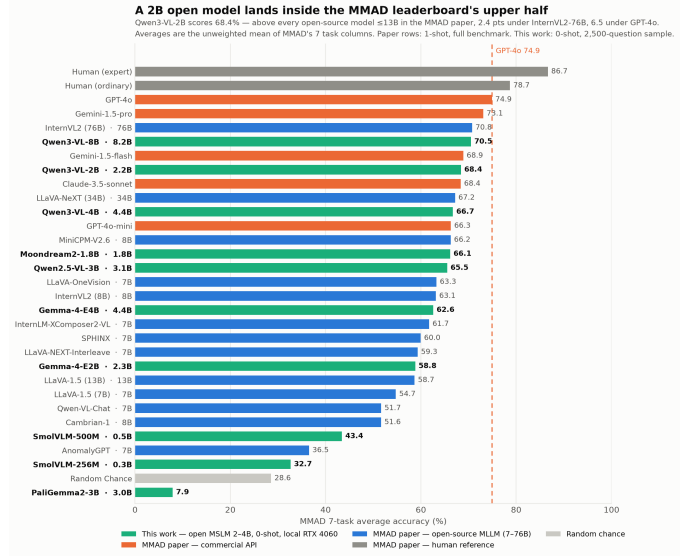


Fig. 3. Our models (green, 0-shot, 2,500 questions) placed on the MMAD leaderboard (1-shot, full benchmark) by MMAD seven-task average.

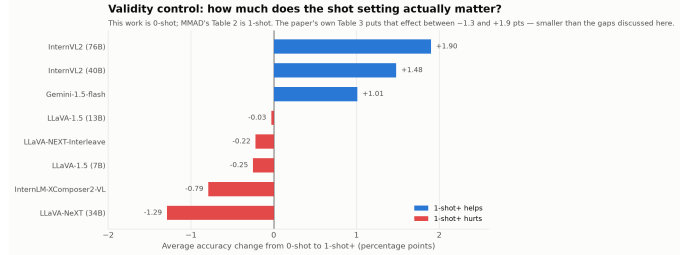


Fig. 4. Validity control from the MMAD paper’s own ablation: moving from 0-shot to 1-shot changes model averages by -1.3 to $+1.9$ points.

severity-4 motion blur and severity-4 Gaussian noise, each also after test-time image restoration (TTA-IR: unsharp masking for blur, bilateral filtering for noise). Table IV and Fig. 7 give the results.

The overall drop is modest ($69.9 \rightarrow 66.1$ under blur, 66.6 under noise), but it hides large opposite movements. Object-level subtasks collapse under blur (Object Classification -19.4 pp, Object Analysis -17.5 pp, Object Structure -13.7 pp), while Defect Classification *rises* by 14.8 pp and Defect Analysis by 9.2 pp. We do not read the rises as robustness: a blurred image is more likely to be judged defective, which helps questions whose premise is that a defect exists. This interpretation still needs a prediction-distribution check [TODO: confusion analysis per subtask].

TTA-IR does not help overall (-0.3 pp on blur, $+0.04$ pp on noise). Its only notable gain is Defect Localization under blur ($+3.4$ pp), within the per-subtask noise band.

Severity sweep (small). A 7-corruption $\times$ 5-severity sweep with 45 questions per cell gave motion blur the steepest slope ($\text{RDS}(5)=1.78$ pper level) but curves that are not monotone in severity; at $n=45$ (± 13 pp) this sweep can rank corruptions only very coarsely and must be repeated at scale.

TABLE III

TEN OPEN MODELS ON THE SAME 2,500 MMAD QUESTIONS (0-SHOT, RTX 4060 LAPTOP). MMAD-7 = UNWEIGHTED MEAN OF MMAD’S SEVEN TASK COLUMNS (BALANCED ANOMALY DISCRIMINATION). DL = DEFECT LOCALIZATION. LATENCY PER QUESTION, BATCH SIZE 1.

Model	Params (B)	Precision	VRAM (GB)	Acc (9 tasks)	κ	MMAD-7	DL	Latency (s)	q/s
Qwen3-VL-8B	8.20	NF4	6.73	72.16	0.627	70.51	55.96	0.32	3.1
Qwen3-VL-2B	2.20	FP16	4.82	71.20	0.615	68.40	47.50	0.16	6.2
Qwen3-VL-4B	4.40	NF4	3.06	68.80	0.582	66.72	51.62	0.19	5.3
Qwen2.5-VL-3B	3.10	FP16	5.86	67.68	0.574	65.51	51.26	0.22	4.5
Moondream2	1.80	FP16	2.10	67.56	0.567	66.13	59.21	0.45	2.2
Gemma-4-E4B	4.40	NF4	3.32	65.60	0.540	62.58	31.41	0.85	1.2
Gemma-4-E2B	2.30	FP16	4.65	60.92	0.476	58.80	26.35	0.34	3.0
SmolVLM-500M	0.50	FP16	1.11	45.72	0.277	43.43	33.21	0.46	2.2
SmolVLM-256M	0.26	FP16	0.85	32.56	0.123	32.66	29.60	0.44	2.3
PaliGemma2-3B (pt)	3.00	FP16	5.80	8.20	−0.076	7.91	10.11	0.21	4.7
MMAD paper (1-shot): GPT-4o / InternVL2-76B / Claude-3.5-sonnet					74.92 / 70.75 / 68.36	55.62 / 56.66 / 48.81	—	—	—

Defect Localization separates vision encoders, not model sizes

Qwen’s dynamic patch models (51.3, 47.5) sit mid-field among the paper’s 7-76B systems, above GPT-4o-mini. Gemma’s pooled-token models (31.4, 26.4) land in the bottom four of the whole table — 6.4 and 1.4 pts off random chance.

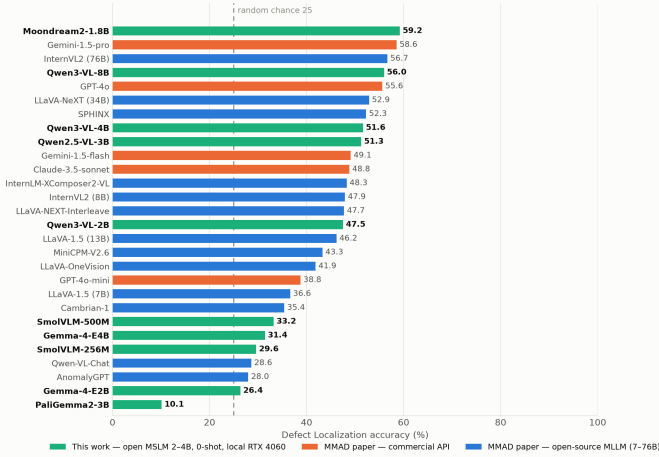


Fig. 5. Defect Localization across all models. Gemma 4 and SmolVLM, which pool visual tokens aggressively, fall near chance. (Figure subtitle predates the 8B and Moondream2 runs [TODO: regenerate caption text in figure].)

Prompt-level mitigation. A defect-anchored reasoning prompt (DAR) reduced accuracy in all six corruption/severity cells of the small mitigation study (−4.4 to −15.6 pp, $n=45$ each), and TTA-IR changed accuracy by 0 to −4.4 pp. The recovery-rate metric used there is undefined when corruption does not lower accuracy, which happened in 5 of 6 cells; it is not reported.

D. First Arm B experiment (Phase 5)

Pipeline. A ResNet-50 PatchCore built a memory bank from up to eight normal images per category, the heatmap was thresholded at a calibrated τ , and the resulting region was drawn as a red box with a “DEFECT CANDIDATE” tag (Fig. 8); the prompt told Qwen3-VL-2B to focus on the red region. If the detector did not fire, the plain image and prompt were used.

Result. On 500 questions, guidance lowered accuracy from 72.2 % to 68.6 % (−3.6 pp). The paired comparison has 38 fixes and 56 breaks, exact McNemar $p=0.079$: the drop is not

TABLE IV

ROBUSTNESS AT SCALE: QWEN3-VL-2B, 5,000 QUESTIONS, SEVERITY 4.

Subtask	Clean	Blur	+TTA	Noise	+TTA
Object Classif.	89.2	69.8	66.3	77.6	78.8
Object Structure	84.3	70.6	70.1	77.5	76.8
Object Analysis	82.5	65.1	62.7	67.6	67.0
Defect Analysis	76.9	86.1	86.9	87.4	87.8
Object Details	74.1	62.3	62.2	67.2	69.0
Defect Descr.	72.6	72.1	71.5	74.1	73.9
Anomaly Det.	55.6	55.2	54.3	53.8	53.8
Defect Local.	47.8	53.2	56.6	50.6	49.6
Defect Classif.	45.8	60.5	61.6	44.1	43.6
All	69.9	66.1	65.8	66.6	66.7

significant, and neither is any per-subtask change ($n \approx 55$ per subtask, ± 13 pp). The pattern in Fig. 9 is nonetheless suggestive: Defect Classification (+5.3 pp) and Defect Analysis (+1.8 pp) improve, while Defect Localization (−10.9 pp) and Object Analysis (−10.7 pp) degrade, consistent with a box that helps name a defect but obscures the object and distorts coarse position judgements.

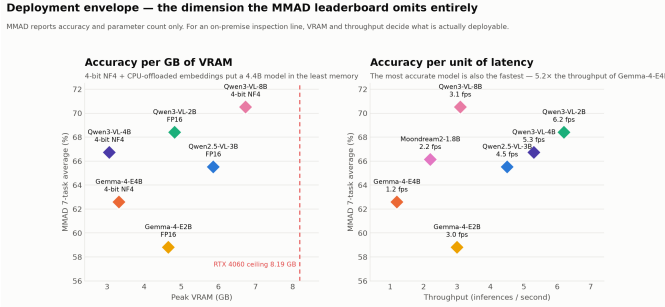
Why this pilot cannot answer the question. Auditing the run revealed three problems:

- 1) *Mis-calibrated detector.* τ was estimated on normal images that were also in the memory bank, so it is biased low. The detector fired on 92.6 % of all images, including normal ones: Arm B was mostly “box everywhere”, not detector guidance.
- 2) *Single-source sample.* All 500 questions came from GoodsAD (the other sources’ normal-image folders did not resolve at the time).
- 3) *Silent parse fallback.* Unparseable outputs defaulted to “A”.

We therefore treat Phase 5 as a negative pilot that motivates the controlled design below rather than as evidence against Arm B.

V. ARM-B LAB: A RESEARCH ENVIRONMENT FOR DETECTOR-GUIDED MLLMS

Answering whether detector guidance helps requires many controlled variants: detectors of known quality, null controls,



Cross-Model Subtask Performance Matrix (MMAD)

Anomaly Detection	61.3	58.6	50.7	47.8	55.8	41.7	59.7	0.0	55.8	45.0
Defect Analysis	82.0	79.6	75.6	33.0	76.7	77.4	69.9	12.5	26.5	76.0
Defect Classification	40.8	59.5	46.9	31.5	48.8	51.2	41.6	2.5	25.4	53.4
Defect Description	71.1	71.1	63.9	40.8	61.7	62.5	69.3	7.2	24.9	65.3
Defect Localization	47.5	56.0	26.4	33.2	51.6	51.3	59.2	10.1	29.6	31.4
Object Analysis	82.3	85.9	71.8	66.8	83.0	83.4	78.7	7.2	33.6	82.7
Object Classification	94.9	90.2	73.3	56.0	91.0	90.6	87.4	13.7	33.6	80.9
Object Details	79.8	72.8	71.7	51.6	76.7	72.0	68.1	9.3	30.1	75.3
Object Structure	81.2	75.8	67.9	50.9	74.0	79.1	74.4	11.2	33.6	80.5

Accuracy (%) scale: 0 to 80

Fig. 6. Left: accuracy against peak VRAM and throughput. Right: subtask accuracy for all ten models (nine MMAD subtask labels).

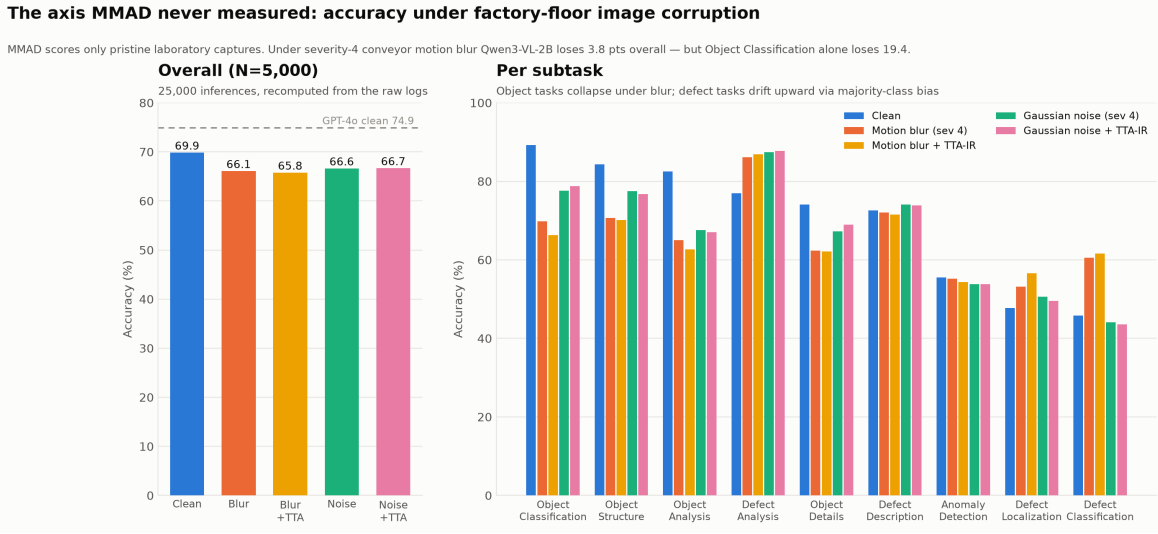


Fig. 7. Clean vs. corrupted vs. restored accuracy, overall and per subtask (25,000 inferences). Object tasks collapse under motion blur; defect tasks drift upward.

prompt styles, gating rules, models. Writing a script per variant is what produced Phase 5’s confounds, so we built *Arm-B Lab* (Fig. 10): a local web application in the style of ComfyUI, in which every block of Fig. 1 is a typed, swappable node.

Architecture. A FastAPI server hosts a node registry; each node declares typed ports (IMAGE, ANOMALY, REGION, QUESTION, RESPONSE, ...) and widget parameters, and the browser builds its palette from that schema. Graphs execute in topological order, and every node output is cached under a fingerprint of (type, parameters, upstream fingerprints), so changing a prompt re-runs only the prompt and the MLLM, and a sweep over a downstream parameter runs the detector once per image. A single worker queue serialises GPU work.

Nodes. Table V lists the node families. Three are designed specifically as experimental controls: a *synthetic detector* built from the ground-truth mask with adjustable miss rate, false-

alarm rate, shift and dilation (with defaults it is an oracle, giving the ceiling of Arm B); a *control detector* that draws a random box, a centre box or the whole image (does *any* box help?); and a *selective gate* that applies guidance only to chosen subtasks, the recommendation that followed from Phase 5.

Evaluation. A batch run takes a stratified MMAD sample and an optional grid of parameter values. Every variant sees the same questions, so arms and variants are compared with paired statistics: accuracy with Wilson intervals, κ , macro accuracy, parse-failure rate, exact McNemar with fixes/breaks, accuracy conditioned on whether the detector fired, RDS and spread across sweep values, and, in letter-logit mode, expected calibration error [24]. Detector nodes are scored against ground-truth masks (image AUROC on τ -normalised scores and per category, pixel AUROC, box IoU, gate TPR/FPR).



Fig. 8. Phase-5 visual prompt: PatchCore region drawn as a red box with a tag.

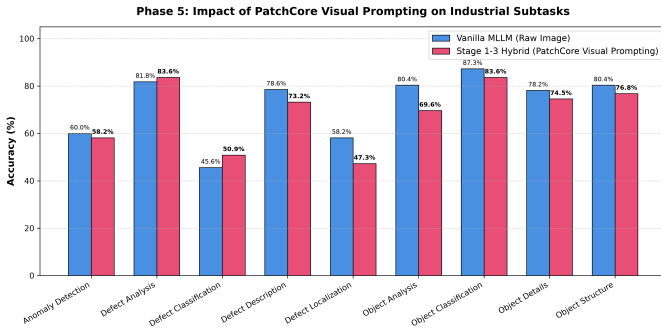


Fig. 9. Phase 5, Arm A (blue) vs. PatchCore-guided Arm B (red), 500 GoodsAD questions. Overall -3.6 pp, McNemar $p=0.079$ (n.s.).

Runs are logged per question as JSONL, are resumable, and export 300-dpi figures and a manifest.

Fixes relative to Phase 5. τ is calibrated on held-out normal images never placed in the memory bank; normal references come from MMAD’s own template lists (query image excluded) for all five sources; parse failures are counted as wrong unless the Phase-5 fallback is explicitly selected for reproduction.

Implementation checks. PatchCore follows [7] (greedy coresets on a random projection, local 3×3 aggregation, configurable backbone and layers). WinCLIP is re-implemented with true window masking on OpenAI ViT-B/16. A first version mixed CLIP’s last-layer patch tokens into the zero-shot map and produced *inverted* maps (pixel AUROC 0.27), consistent with the known misalignment of those tokens [25]; the corrected version uses window embeddings only for zero-shot and penultimate-layer tokens for the few-shot memory. On a 24-image check (Table VI) the expected ordering PatchCore > WinCLIP+ > WinCLIP zero-shot holds. On the GPU, Qwen3-VL-2B answers in 0.17 s per question (4.5 GB) in generation mode and 0.18 s in letter-logit mode.

TABLE V
ARM-B LAB NODE FAMILIES.

Family	Nodes (key parameters)
Input	MMAD sample (source, subtask, defective/normal); normal references (k , template strategy); own image / question
Pre-process	Corruption engine (7×5); test-time restoration
Question	Question builder (3 paraphrases, option shuffling)
Detector	PatchCore (backbone, layers, resolution, coreset, k -NN, τ margin); WinCLIP / WinCLIP+ (CLIP model, scales, templates); synthetic GT detector; null control
Localise	Heatmap $\rightarrow$ box (gate: calibrated/fixed/always/never; threshold: calibrated/relative/percentile/Otsu)
Prompt	Visual prompt (9 styles: box, box+wash, contour, heatmap, crop, side-by-side, blur/darken outside); text prompt (7 templates incl. hedged hint); selective gate
Reason	MLLM (any local Qwen/SmolVLM/Gemma, precision, generate vs. letter-logits, temperature, seed)
Evaluate	Answer parser (strict/lenient/tag); score; detector metrics

TABLE VI
DETECTOR SANITY CHECK IN THE LAB (24 VISA/DS-MVTEC IMAGES, 12 DEFECTIVE, $k=8$ NORMAL REFERENCES; IMAGE AUROC ON RAW SCORES POOLED ACROSS CATEGORIES). PRELIMINARY — TO BE REPLACED BY THE FULL DETECTOR BENCHMARK (E3).

Detector	Image AUROC	Pixel AUROC
PatchCore (WRN-50, layer2+3)	0.861	0.923
WinCLIP+ (few-shot, $k=8$)	0.833	0.825
WinCLIP (zero-shot)	0.792	0.684

VI. DISCUSSION

A. What the completed phases show

(i) A 2B open model on a laptop GPU is a credible zero-shot inspector on MMAD’s question types, within about 6.5 points of GPT-4o’s 1-shot average; (ii) the weak spots are fine-grained defect classification and localisation, exactly where a pixel-level detector should help; (iii) aggregate accuracy understates corruption damage, and training-free restoration does not repair it; (iv) naive detector-guided prompting did not help, but the only test of it was confounded.

B. Threats to validity

Shot setting: our zero-shot numbers are compared with MMAD’s 1-shot table. *Sample size:* 2,500 of 39,672 questions; per-subtask differences under ~ 6 pp are not resolvable. *Question sets:* Phase 1 and Phase 4 draws differ slightly. *Quantisation:* 8B, 4B and Gemma-4-E4B ran in NF4, the others in FP16, so architecture and precision are confounded. *Model vintage:* our models are one to two years newer than the paper’s. *Label inference:* the normal/abnormal split for balanced accuracy is inferred from the `/good/` path convention. *Single seed:* all runs are greedy with one sampling seed.

C. Consistency audit of the earlier manuscript

Recomputing every table from the raw logs showed that 28 of 36 per-subtask cells in the earlier thesis draft disagree with the artefacts by more than 0.5 points. The draft’s “Phase 1” table matches the clean column of the 5,000-question study

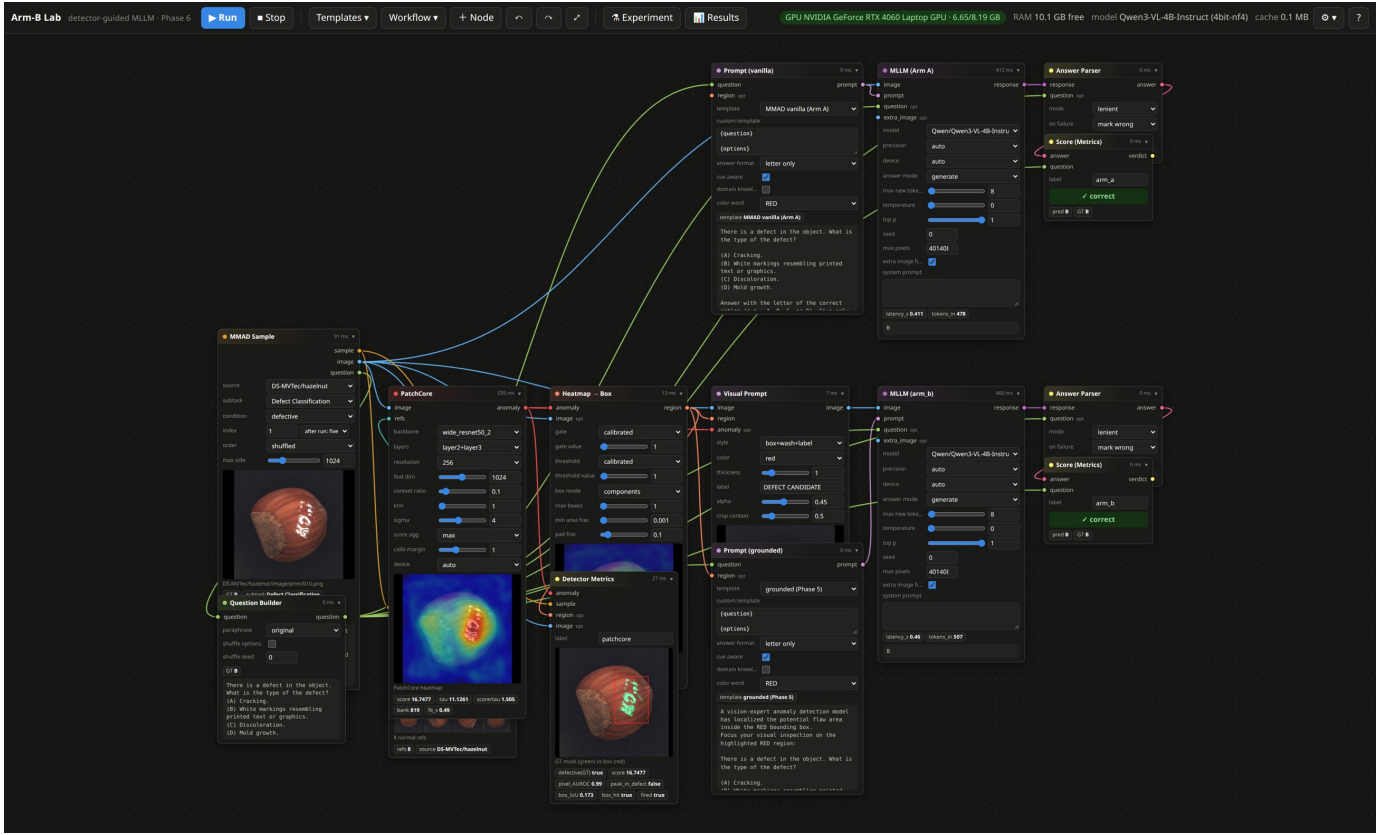


Fig. 10. Arm-B Lab running the paired Arm A (top row) vs. Arm B (bottom row) graph on an MMAD hazelnut “print” defect with Qwen3-VL-4B (NF4). Each node shows its output: PatchCore heatmap (score/ τ =1.51), the box drawn for the MLLM, the ground-truth mask vs. box (pixel AUROC 0.99, box IoU 0.17), the prompts and both verdicts. Right-click replaces PatchCore by WinCLIP or a control detector while keeping the wiring.

(mean absolute difference 0.08) rather than the Phase-1 run (3.62), i.e. it is correct data under the wrong label; the cross-model subtask table diverges by up to 11.7 points and must be regenerated. All numbers in this draft come from the artefacts.

VII. REMAINING WORK

Table VII lists the experiments planned with Arm-B Lab. Each is paired (same questions across arms) and uses all five MMAD sources. At $N=1,800$ (200 per subtask) the overall 95 % half-width is about ± 2.1 pp; paired McNemar tests are more sensitive than that. With Qwen3-VL-2B at ~ 0.2 s per question, the MLLM cost of one arm at $N=1,800$ is about six minutes.

Writing tasks: regenerate the manuscript tables from the manifests; replace Phase 2’s small severity sweep by E9 at scale; confirm the majority-class interpretation of the corruption results with confusion matrices; verify the TODO-marked references.

VIII. CONCLUSION

Small open MLLMs running on a single laptop GPU already answer industrial-inspection questions at a level close to large commercial models, but they are weakest at naming and locating defects and are fragile to motion blur in ways that aggregate accuracy hides. Our first attempt at detector

guidance did not help, for reasons we can now identify and control. Arm-B Lab turns the remaining question, when a classical anomaly detector helps a small MLLM and when it misleads it, into a set of paired, reproducible experiments, which form the next stage of this work.

REPRODUCIBILITY

Code, run logs and figures are in the Open-IAD repository: phases 0–5 under Local-training/phase-*, the re-computed comparison under comparative-analysis/, and Arm-B Lab under Local-training/phase-6 (./run_lab.sh).

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TABLE VII
PLANNED EXPERIMENTS (ALL RUNNABLE AS SAVED ARM-B LAB TEMPLATES).

ID	Experiment	Question it answers	Design
E1	Arm A vs. calibrated PatchCore	Does detector guidance help once the detector is calibrated and data are balanced?	$N=1,800$, 5 sources, McNemar, acc. conditioned on detector firing
E2	Oracle / random box / no box	Ceiling of Arm B; does <i>any</i> box help?	Synthetic GT detector vs. control detector vs. Arm A, defective images
E3	Detector benchmark	PatchCore vs. WinCLIP(+) quality on MMAD images	Image/pixel AUROC, IoU, gate TPR/FPR, one row per image
E4	Detector-quality sweep	How good must the detector be?	Sweep synthetic miss rate, false-alarm rate and shift
E5	Gate & threshold sweep	Precision/recall trade-off of firing	Sweep gate value and box threshold
E6	Visual-prompt style	Box vs. contour vs. crop vs. heatmap vs. blur-outside	Same detector, paired across styles
E7	Text prompt / trust	Does a hedged hint avoid the harm of false boxes?	Grounded vs. hedged vs. score-annotated prompts
E8	Selective gating	Guide only Defect Classification/Analysis?	Gated vs. always-guided vs. Arm A
E9	Robust Arm B	Does the detector help or fail under corruption?	Severity 0–5 sweep on both arms, RDS per arm
E10	Prompt sensitivity	Paraphrase spread, letter-position bias	4 paraphrases $\times$ option shuffling
E11	Calibration	Are confident answers correct?	Letter-logit probabilities, ECE, reliability diagrams
E12	Model scaling of Arm B	Does guidance matter less for 8B than for 2B?	Qwen3-VL 2B/4B/8B, Moondream2, Gemma-4
E13	Official protocol	Full-benchmark, 1-shot numbers	MMAD code on all 8,366 images incl. GoodsAD
E14	Repeatability	Seed and run-to-run variance	3 sampling seeds for the headline comparisons

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